SELVARANJITH P

Biomedical Engineering Student

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CAREER OBJECTIVE

Biomedical Engineering student with a diploma background in Health/Sanitary Inspection, seeking an entry-level opportunity to apply technical knowledge of medical equipment, healthcare systems, and analytical problem-solving skills in a biomedical engineering role.

EDUCATION

Biomedical Engineering — *Currently Pursuing* | *CGPA: 7.78*

Health Inspector (HI/SI) | Subhasakthi Institute of Paramedical Science, Sathiyamangalam — *73%* | *2023*

HSLC | Mullai Higher Secondary School, Thottiyam — *63%* | *2020*

SSLC | Government High School, Thalamalaipatti — *75%* | *2018*

TECHNICAL SKILLS

- Programming Languages: C++, Java
- Web: HTML

SOFT SKILLS

- Communication
- Teamwork
- Problem-Solving
- Leadership

INTERNSHIP EXPERIENCE

Galwin Technology | Trichy

- Completed hands-on training in embedded systems for biomedical applications, gaining practical experience with microcontrollers, sensors, and medical device interfacing.
- Developed foundational skills in embedded hardware, circuit design, sensor integration, and programming for healthcare technology applications.

Newtown Biomedical Service | Karur

- Gained hands-on experience in the handling, installation, and preventive maintenance of diagnostic and laboratory medical equipment.
- Performed basic troubleshooting, fault identification, and servicing of biomedical instruments to ensure reliable equipment performance and patient safety.

Hospital Training - Harsha Mitra | Trichy

- Completed hospital training with hands-on exposure to clinical workflows, biomedical equipment, and healthcare technology used in patient care.
- Observed the operation, preventive maintenance, troubleshooting, and safety protocols of medical devices across various hospital departments.

PROJECTS

Mini Project - II: MEMS-Based Smart Goggles for Early Glaucoma Risk Detection

- Designed a wearable smart goggle prototype using an ESP32 and MEMS pressure sensor for non-invasive intraocular pressure (IOP) monitoring.

- Developed a real-time monitoring system with sensor data processing and alert functionality for early glaucoma risk detection.

Mini Project - I: AI-Based Swallowing Tracking Pattern for Dysphagia

- Developed an AI-based system to monitor and analyze swallowing patterns for the early detection of dysphagia.
- Integrated sensor data processing with machine learning techniques to improve swallowing assessment and support timely clinical intervention.

CERTIFICATIONS

- Core Java - Yellow Matrix
- Medical Coding - Thoughts Flow
- Biomedical Hardware Training - Atheenapandian
- NPTEL - Internet of Things (Elite, 71%)

ACHIEVEMENTS & HACKATHONS

- Runner-Up, National Level Hackathon - India-Israel Hackathon 2026, for developing a "Capacitive Deionization-Based Water Purification System".
- Shortlisted among 159 teams at EYEQINOVATE 2.0 for presenting GlaucoScan before ophthalmologists and StartupTN evaluators.
- Won Second Prize, Inter-College Hackathon - Electroblitz 2025, for building an "Automatic Medicine Dispenser System".